

Economic and Ecological Losses from Wildfires using Machine Learning

Master Thesis Defense

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Conceptual visual: wildfire hazard + mapped risk + consequence protection

Presentation Roadmap

From research gap to defended contribution.



01 Motivation & gap

Why wildfire ML must move beyond hazard prediction

02 Methodology & data

FEMA/NRI EAL, county predictors, environmental augmentation

03 Experimental results

Baseline, environmental gain, feature importance, grouped validation

04 Risk outputs

Predicted loss ranking and ecological screening

05 Supporting work & appendices

Vietnam study, Creek Fire workflow, certificate, limitations

Motivation: From Fire Prediction to Consequence Assessment

The thesis focuses on losses, not only whether or where fires occur.

Core problem

- Wildfire ML often predicts occurrence, spread, susceptibility, or burned area.
- Defense-relevant decisions require consequence-oriented outputs.
- This thesis estimates monetary wildfire risk and adds ecological exposure screening.

Research direction

- Public data → reproducible national-scale baseline
- Environmental drivers → better wildfire-specific signal
- Grouped validation → more realistic generalization test



Research Gap, Novelty, and Main Contribution

The novelty is a transferable consequence-oriented workflow, not a territory-specific result.

Unlike much previous wildfire ML work

- Does not stop at hazard, susceptibility, or burned-area prediction.
- Connects environmental fire drivers with exposure, vulnerability, resilience, and monetary EAL.
- Adds ecological screening without claiming direct measured ecological loss.

New contribution / insights

- End-to-end FEMA/NRI-based wildfire-loss pipeline.
- Environmental augmentation produces the strongest performance gain.
- State-wise grouped validation strengthens the defense by testing geographic transferability.
- Risk outputs include predicted EAL, ranking, error analysis, and ecological overlap.

Literature Review: What the Thesis Builds On

Chapter 2 connects wildfire science, ML prediction, loss modeling, ecological vulnerability, and emerging AI.



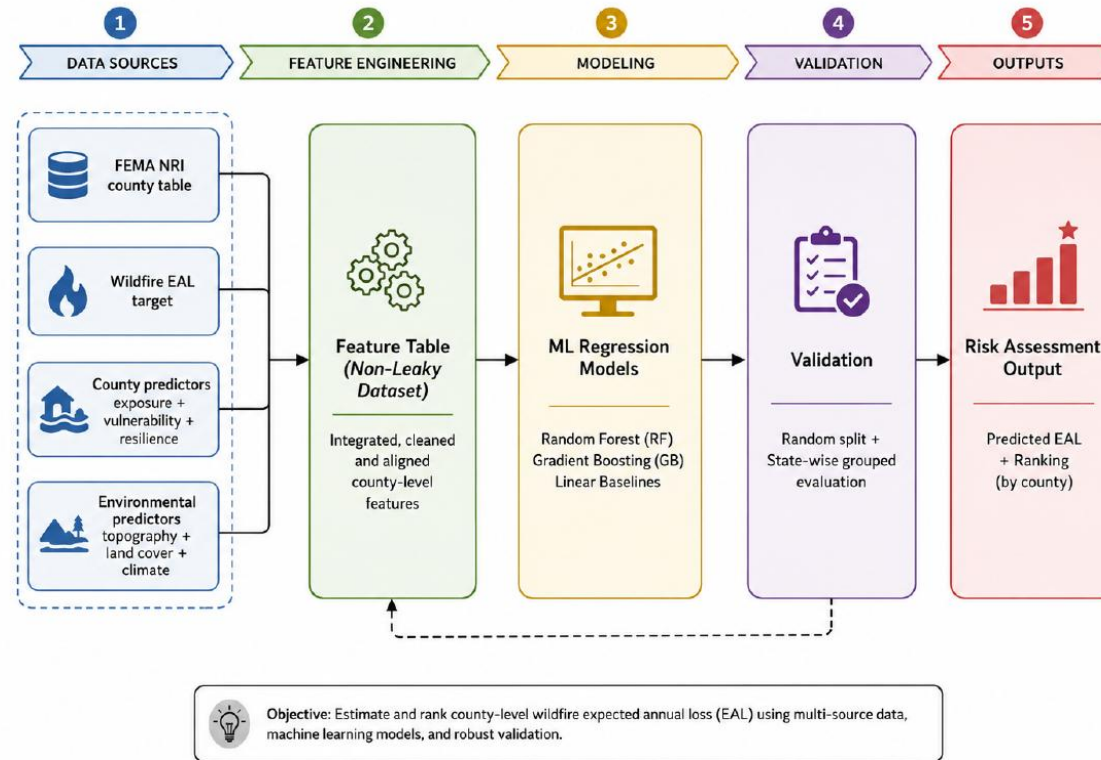
Key synthesis: wildfire-loss assessment needs hazard-relevant environmental context plus exposure, vulnerability, and validation beyond random splits.

Selected sources from Thesis Chapter 2: Rothermel (1972); Chuvieco et al. (2010); Ager et al. (2010); Balch et al. (2017); Roberts et al. (2017); Valavi et al. (2019); Meyer & Pebesma (2021).

Methodology: End-to-End FEMA/NRI Pipeline

The final thesis pipeline turns public county and environmental data into risk-assessment outputs.

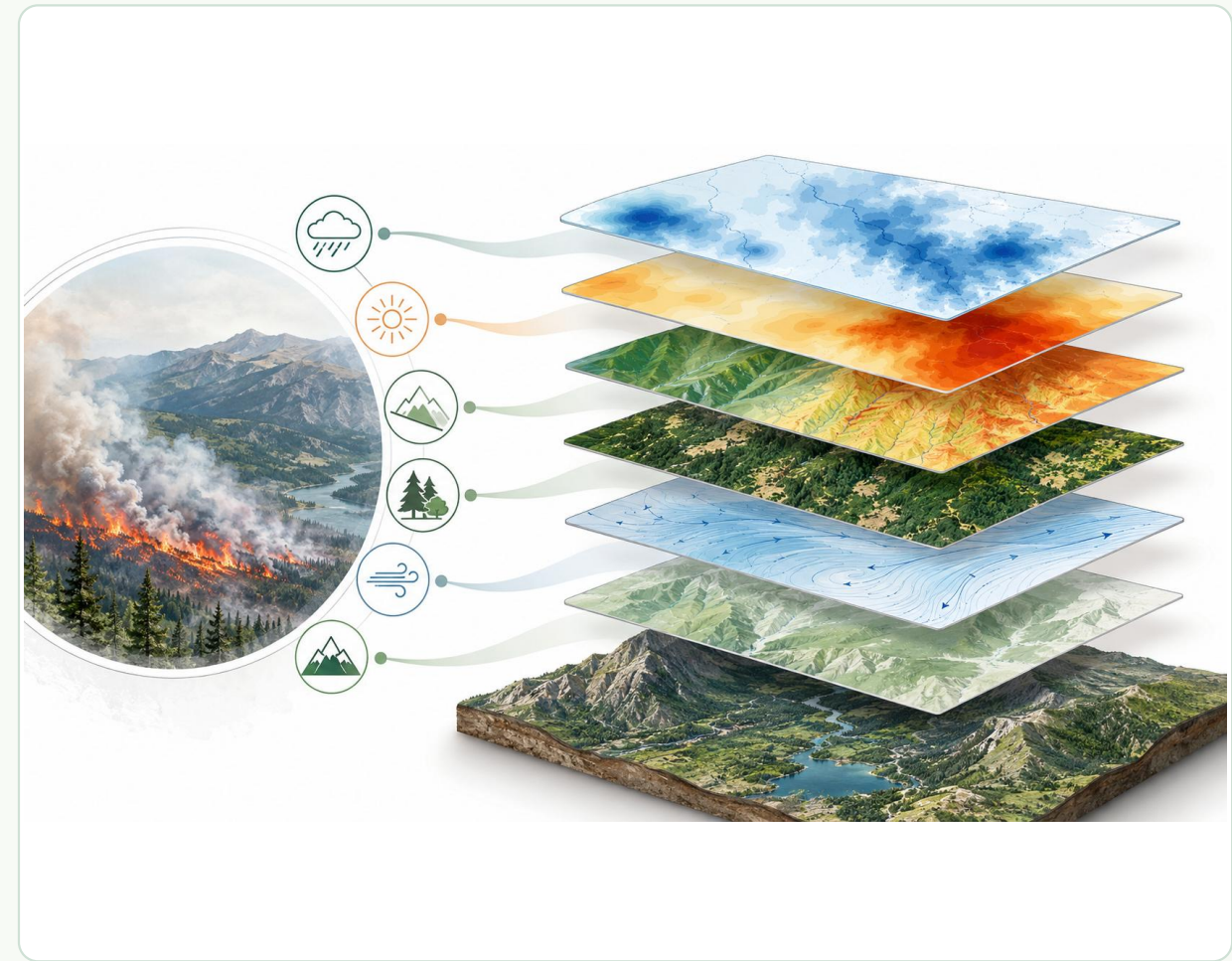
Public loss data → predictor engineering → machine-learning models → geographic validation → economic and ecological screening outputs



Data Foundation: From Risk Index to Environmental Context



The model combines public loss indicators with exposure, vulnerability, land cover, topography, and climate.

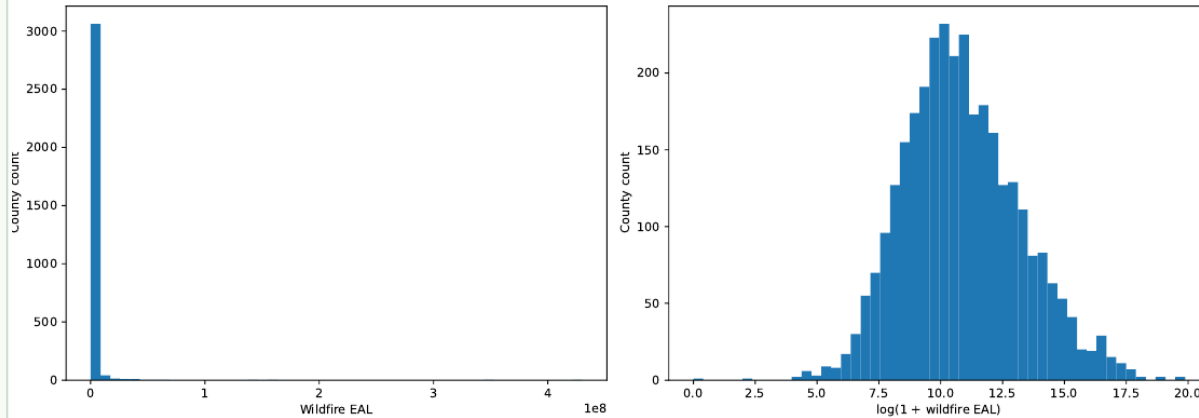


Layer	Examples	Purpose
Target	Wildfire EAL	monetary risk
Exposure/value	population, building, agriculture	assets at risk
Vulnerability	SOVI, RESL	social capacity
Terrain	elevation, slope	fire behavior
Land cover	forest, shrub, grass	fuel/exposure
Climate	T, precipitation, wind, VPD	fire weather

Source: Thesis Chapter 3, Sections 3.4–3.7; conceptual image used only to illustrate data layers.

Target Variable: Wildfire Expected Annual Loss

The thesis predicts a monetary risk indicator from FEMA NRI, transformed for regression.



Target: wildfire Expected Annual Loss (EAL)

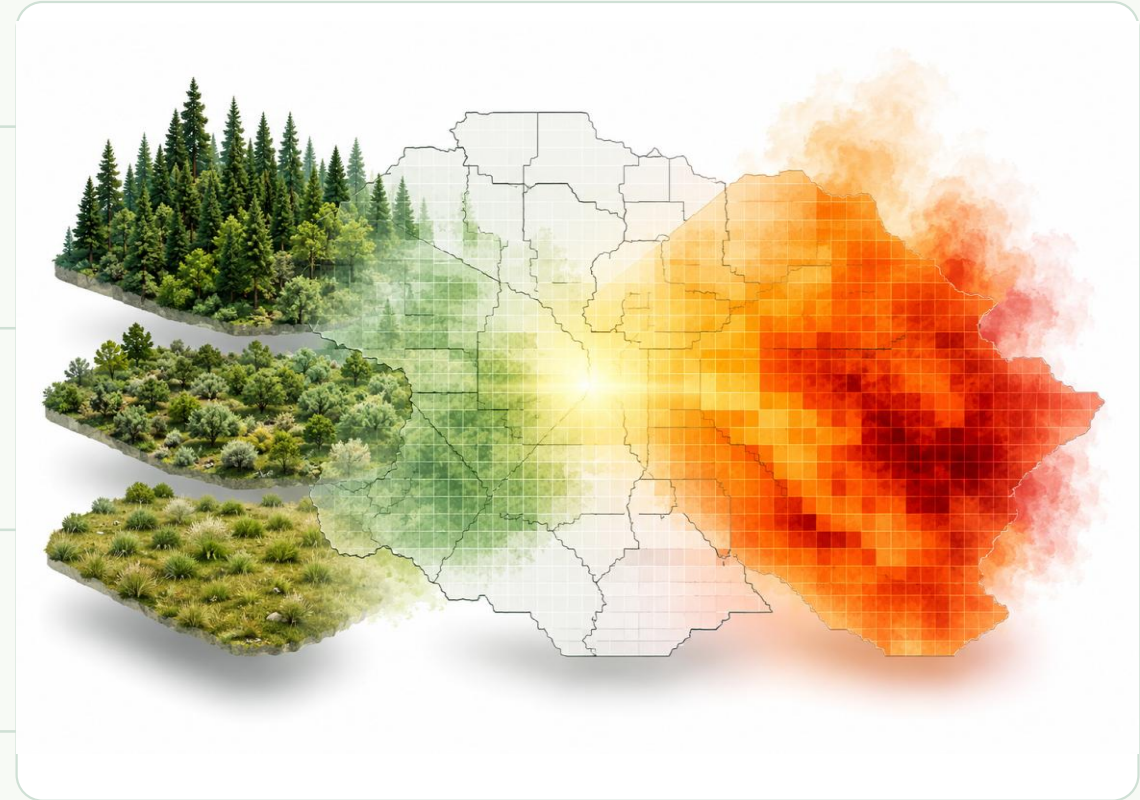
Modeling target: $\log(1 + \text{wildfire EAL})$

- Original county dataset: 3,232 geographic units.
- Cleaned target table: 3,144 counties after removing missing wildfire target values.
- Log transformation reduces extreme skew and stabilizes the regression target.

Modeling and Validation Strategy

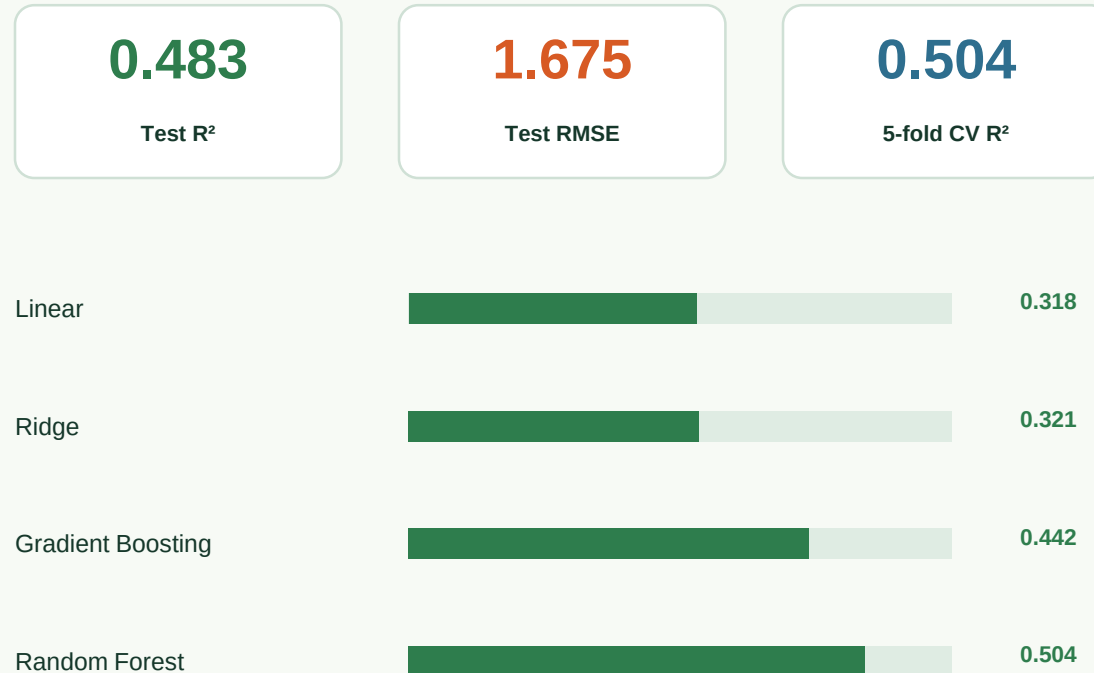
The design compares interpretable baselines, nonlinear ensembles, and stricter spatial validation.

- 1 County-only baseline**
Exposure, value, vulnerability, resilience
- 2 Environmental augmentation**
Topography + land cover + climate
- 3 Model comparison**
Linear, Ridge, RF, Gradient Boosting
- 4 Validation**
Random split + 5-fold CV + state-wise grouped CV



Baseline County-Level Model: Meaningful, but Limited

County-only predictors provide signal, but leave substantial variation unexplained.



Interpretation

- Random Forest performs best among county-only models.
- Nonlinear structure matters: simple linear baselines are weaker.
- Baseline variables alone cannot fully capture wildfire hazard context.
- RMSE values are calculated on the **log-transformed wildfire EAL scale**.

Why this matters

- The baseline establishes a defensible starting point before environmental augmentation.

Environmental Augmentation: Major Performance Gain

Adding topography, land cover, and climate turns the baseline into a strong loss predictor.

Table 4.3: Model comparison after adding environmental predictors to the county-level baseline dataset.

Feature set	Model	MAE	RMSE	R^2	CV R^2	CV RMSE
Combined baseline + environment	Random Forest	0.779	1.011	0.829	0.795 ± 0.022	1.049 ± 0.023
Combined baseline + environment	Gradient Boosting	0.864	1.085	0.804	0.776 ± 0.019	1.099 ± 0.018
Environmental only	Random Forest	0.817	1.112	0.794	0.769 ± 0.017	1.118 ± 0.030
Environmental only	Gradient Boosting	0.970	1.255	0.737	0.713 ± 0.018	1.245 ± 0.034
Baseline (CONUS)	Random Forest	1.287	1.631	0.556	0.520 ± 0.028	1.609 ± 0.027
Baseline (CONUS)	Gradient Boosting	1.316	1.649	0.546	0.498 ± 0.037	1.646 ± 0.046

0.829

Combined RF test R^2

0.795

Combined RF CV R^2

1.011

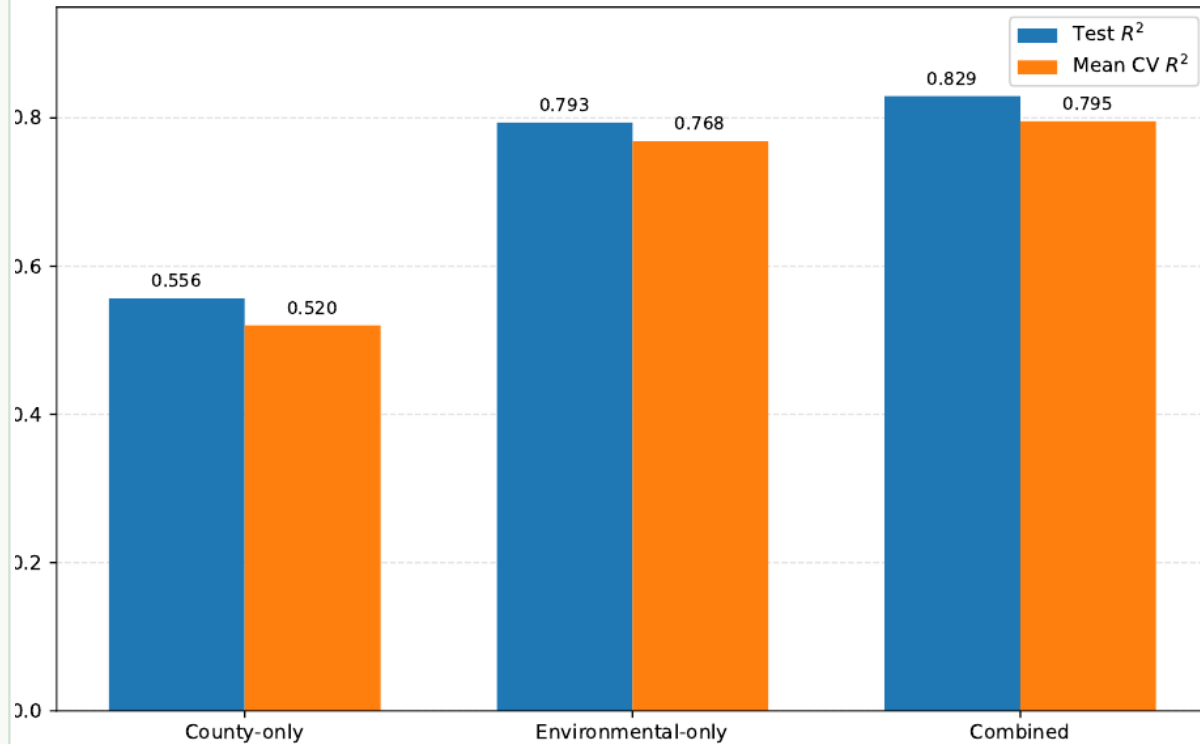
Combined RF RMSE

- Combined baseline + environment is strongest.
- Environmental-only model already performs strongly.
- This supports the thesis claim that hazard-relevant environmental context is essential for loss modeling.

Source: Thesis Chapter 4, Table 4.3.

Performance Gain: Environment Carries Transferable Signal

The strongest improvement appears when environmental features are added.



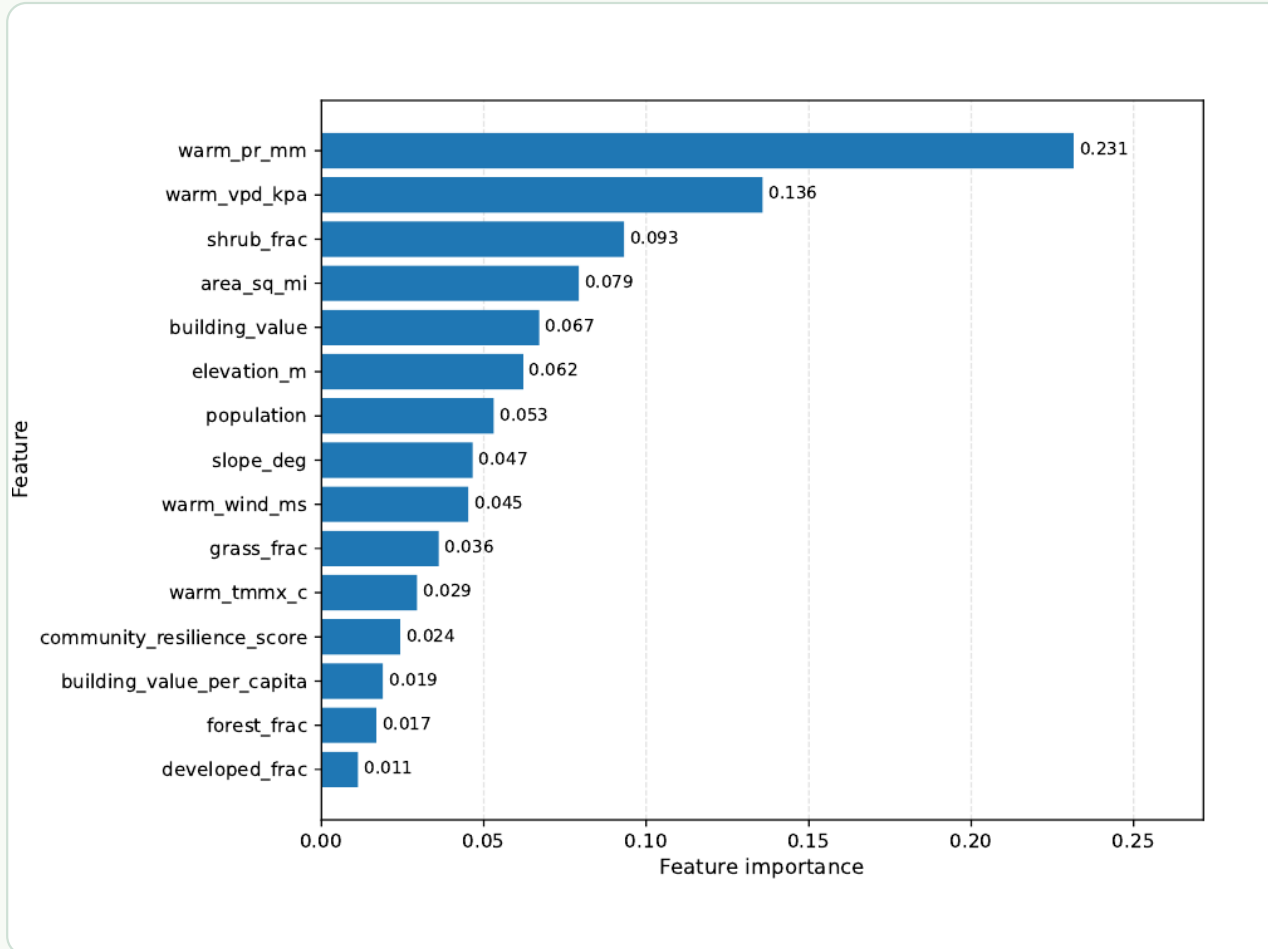
Main insight

- County-only baseline: useful but incomplete.
- Environmental-only model: strong signal from hazard context.
- Combined model: best overall, joining exposure with environmental conditions.
- **Random 5-fold CV result; state-wise grouped validation is reported separately.**

Practical meaning: loss prediction improves when the model sees where fuels, climate, and terrain make wildfire losses plausible.

Feature Importance: Climate and Vegetation Dominate

The combined model learns a wildfire-specific environmental risk signal.



Most influential predictors

- Warm-season precipitation
- Vapor pressure deficit
- Shrub fraction
- County area (sq. mi.)
- Elevation / slope
- Building value / population

Feature labels:

warm_pr_mm = warm-season precipitation

warm_vpd_kpa = warm-season vapor pressure deficit

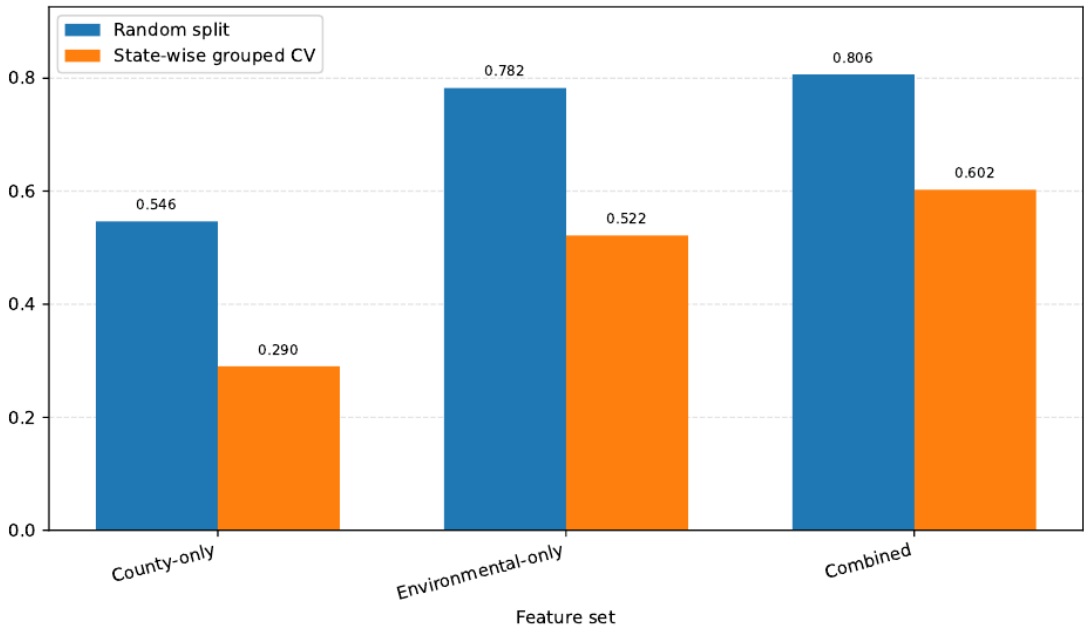
area_sq_mi = county area in square miles

Interpretation: the model is not relying only on socioeconomic proxies; environmental hazard variables dominate the strongest signals.

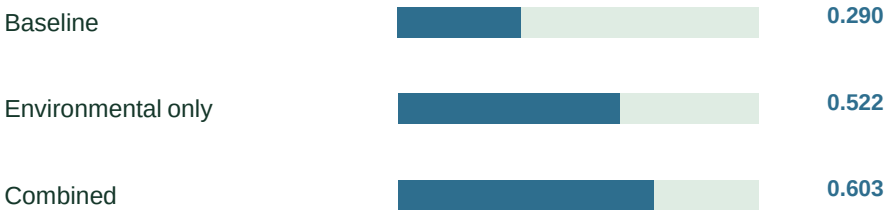
State-Wise Geographic Validation: Stronger Test

Grouped validation avoids overly optimistic random geographic mixing.

Feature set	<i>n</i> features	Random R^2	Random RMSE	Random MAE	Grouped CV R^2	Grouped CV RMSE	Grouped CV MAE
Combined baseline + environment	20	0.806	1.069	0.809	0.603 ± 0.041	1.404 ± 0.155	1.080 ± 0.113
Environmental only	10	0.782	1.133	0.830	0.522 ± 0.074	1.534 ± 0.164	1.169 ± 0.117
Baseline (CONUS)	10	0.546	1.636	1.284	0.290 ± 0.101	1.863 ± 0.109	1.486 ± 0.098



Random vs. grouped R^2

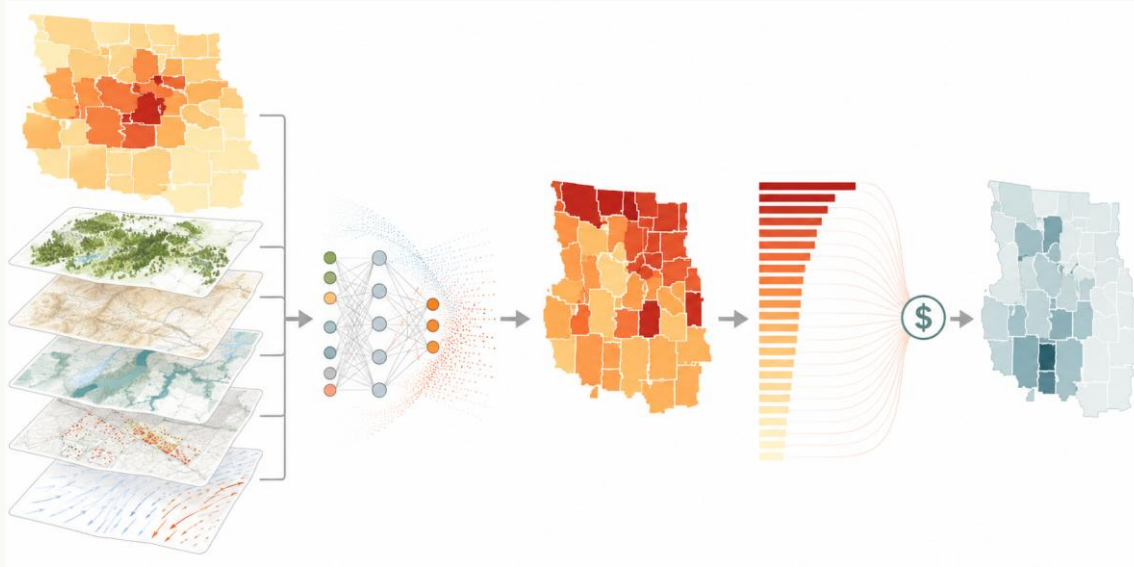


- Performance decreases under grouped validation, as expected.
- Combined model remains best under the stricter test.
- Environmental predictors provide the strongest transferable signal across states.
- **The lower grouped R^2 is expected because entire states are held out during validation.**

Source: Thesis Chapter 4, Table 4.5 and Figure 4.9.

From Model Score to Risk-Assessment Output

The final pipeline produces interpretable county-level outputs, not only model metrics.



Predicted EAL

monetary estimate

Risk ranking

county prioritization

Error analysis

under/overprediction

Ecological screen

overlap signal

Inverse transform: predicted EAL = $\exp(\text{predicted log EAL}) - 1$

Rank	County	State	Predicted Wildfire EAL
1	Riverside	CA	\$146.9M
2	San Diego	CA	\$127.4M
3	Los Angeles	CA	\$123.3M
4	San Bernardino	CA	\$75.3M
5	Orange	CA	\$73.9M

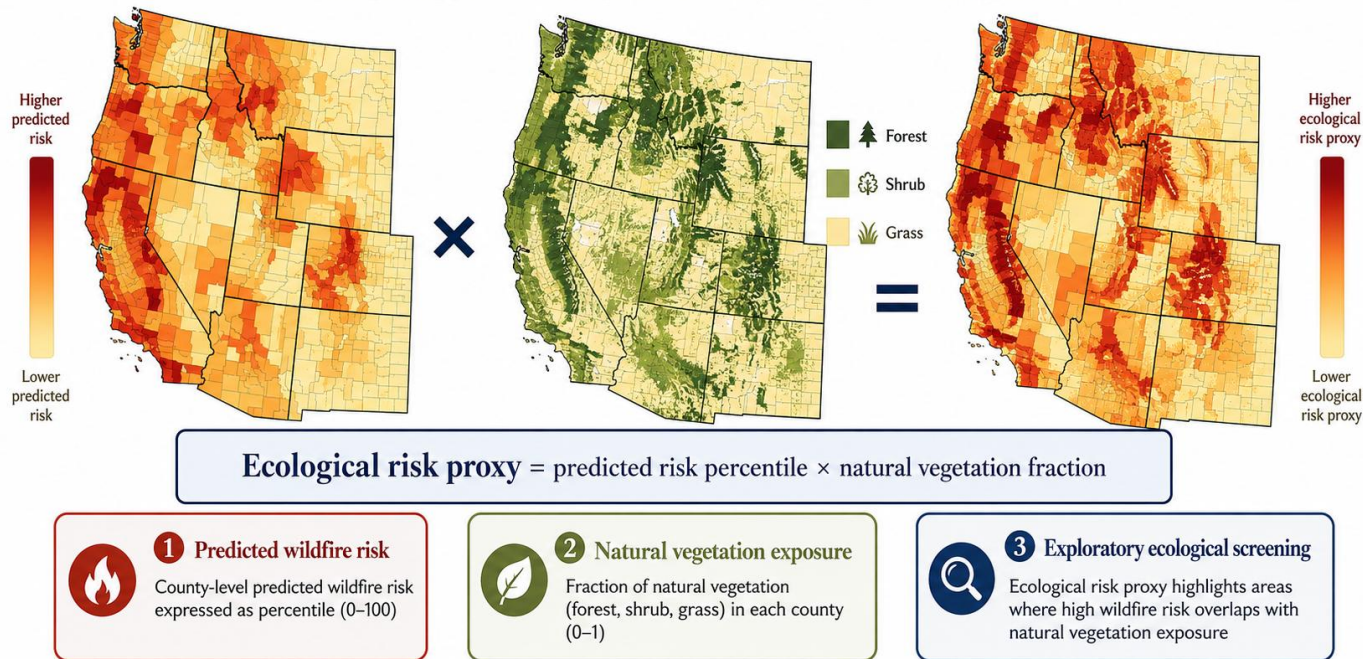
Source: Thesis Chapter 4, Section 4.11, Table 4.6; conceptual image illustrates output flow.

Exploratory Ecological Risk Screening

The ecological layer is a proxy: it does not directly estimate measured ecological loss.

EXPLORATORY ECOLOGICAL RISK SCREENING IN WILDFIRE ANALYSIS

Integrating predicted wildfire risk with natural vegetation exposure



What the proxy captures

- High predicted monetary wildfire risk
- High natural vegetation exposure
- Overlap signal for further ecological vulnerability or recovery analysis

Predicted wildfire-risk percentile = rank of predicted wildfire EAL among all counties, scaled from 0 to 100.

What it does not claim

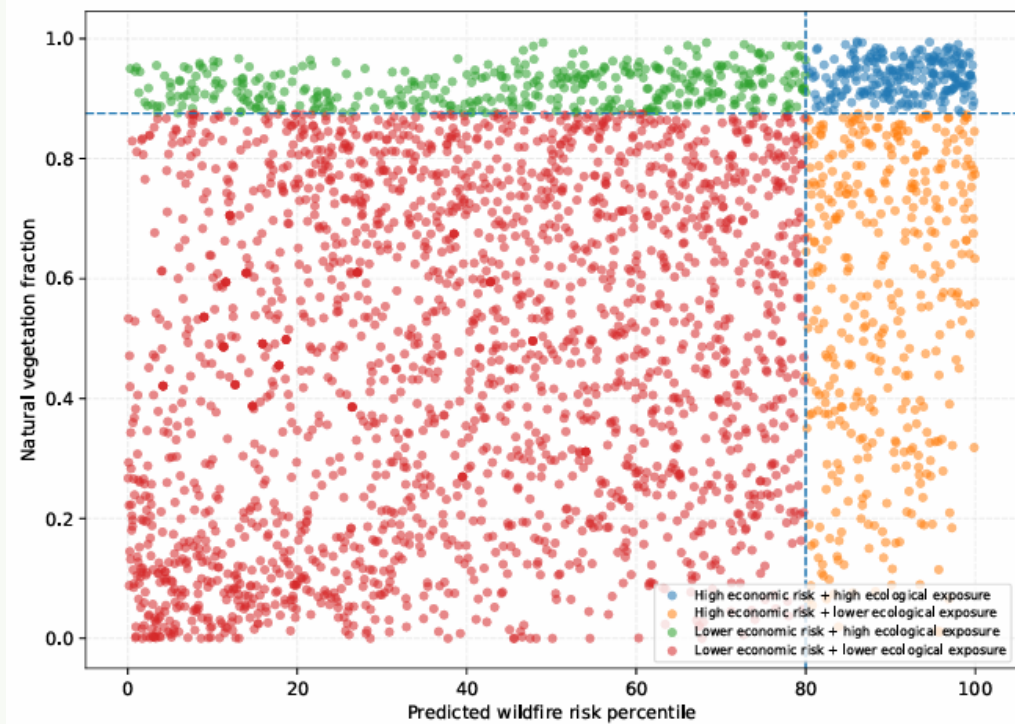
- It is not measured ecological loss.
- It is a screening layer for prioritization and future work.

Ecological risk proxy = predicted wildfire-risk percentile × **natural vegetation fraction**

Natural vegetation fraction = forest + shrub + grass fractions

Economic–Ecological Overlap: Practical Screening Output

The output separates high monetary risk from high ecological exposure.



- Counties in the high-high quadrant combine predicted wildfire EAL risk with high natural vegetation exposure.
- This creates a defensible bridge from monetary loss modelling to ecological consequence screening.
- The Appendix A.7 preserves the component values for forest, shrub, and grass fractions.

238 counties

High economic risk
+
high ecological
exposure

85.62

Mean ecological
risk proxy

0.94

Mean natural
vegetation fraction

\$3.47M

Mean predicted
wildfire EAL

Risk-overlap quadrant	County count	Mean observed wildfire EAL	Mean predicted wildfire EAL	Mean predicted risk percentile	Mean natural vegetation fraction	Mean ecological risk proxy
High economic risk + high ecological exposure	238	5,315,258	3,466,650	91.37	0.94	85.62
High economic risk + lower ecological exposure	395	6,047,578	3,719,458	89.19	0.55	49.63
Lower economic risk + high ecological exposure	395	88,208	66,091	44.94	0.92	41.48
Lower economic risk + lower ecological exposure	2,135	59,641	49,950	39.10	0.46	19.47

Summary of Main Experimental Results

The strongest setup is the combined baseline + environmental Random Forest.

Three takeaways

- County-level wildfire EAL can be modeled with meaningful predictive power.
- Environmental predictors contribute the strongest performance gain.
- Grouped validation shows useful, but not perfect, geographic transferability.

0.829

Best random-split test
 R^2

0.795

Best random CV R^2

0.603

State-wise grouped R^2

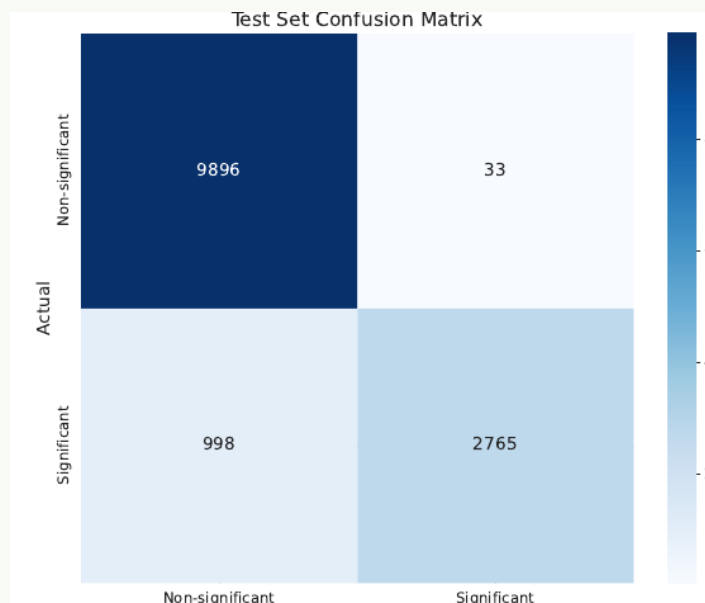
Model setup	Best model	Test R^2	CV / grouped R^2	Notes
County-only baseline	Random Forest	0.483	0.504	Accessible county variables only
Environmental-only (CONUS)	Random Forest	0.793	0.768	Topography, land cover, climate
Combined baseline + environment (CONUS)	Random Forest	0.829	0.795	Standard random cross-validation
Combined baseline + environment (state-wise)	Random Forest	0.806	0.602	Grouped cross-validation by state

Source: Thesis Chapter 4, Table 4.12.

Supporting Study: An Giang, Vietnam Wildfire Prediction

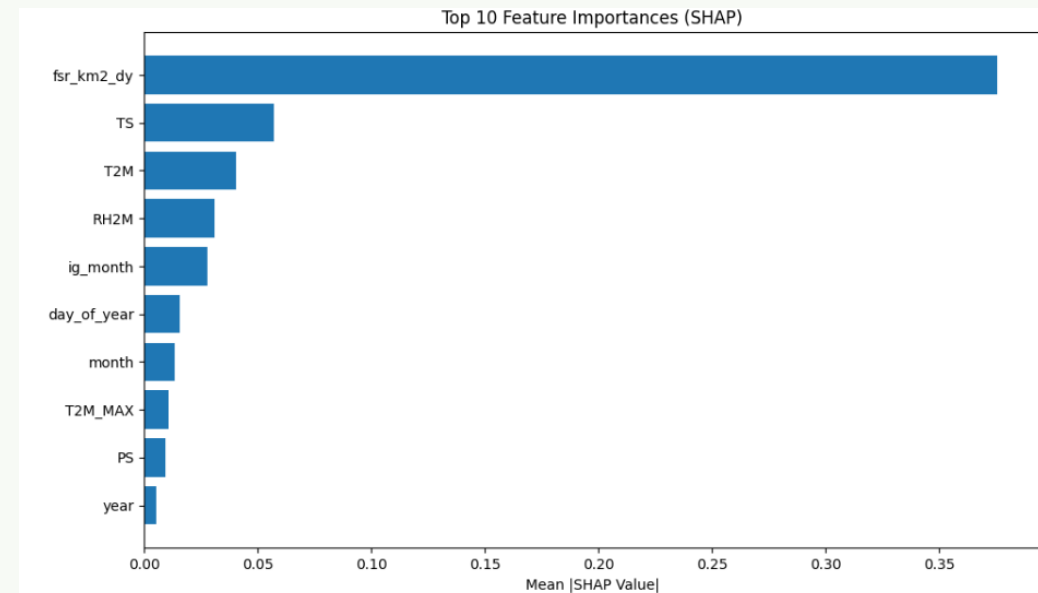


Included as supporting work; the main thesis contribution remains the FEMA/NRI loss-assessment pipeline.



Why it belongs in the defense

- Demonstrates earlier fine-scale deep-learning work on wildfire prediction.
- Uses environmental, terrain, vegetation, and spread-related predictors.
- Provides SHAP-based interpretability and conference-presented supporting evidence.
- Kept to one slide so the main defense remains focused on the thesis pipeline.



Connection: hazard-prediction experience → thesis-level loss-assessment framework

Metric	Value
Overall accuracy	91.4%
Significant fire recall	74.8%
Significant fire precision	92.5%
F1-score	82.7%
Specificity	97.7%
False positive rate	2.3%
False negative rate	25.2%
Dominant SHAP feature	Fire spread rate
Fire spread rate importance	37.6%
Optimal decision threshold	0.65

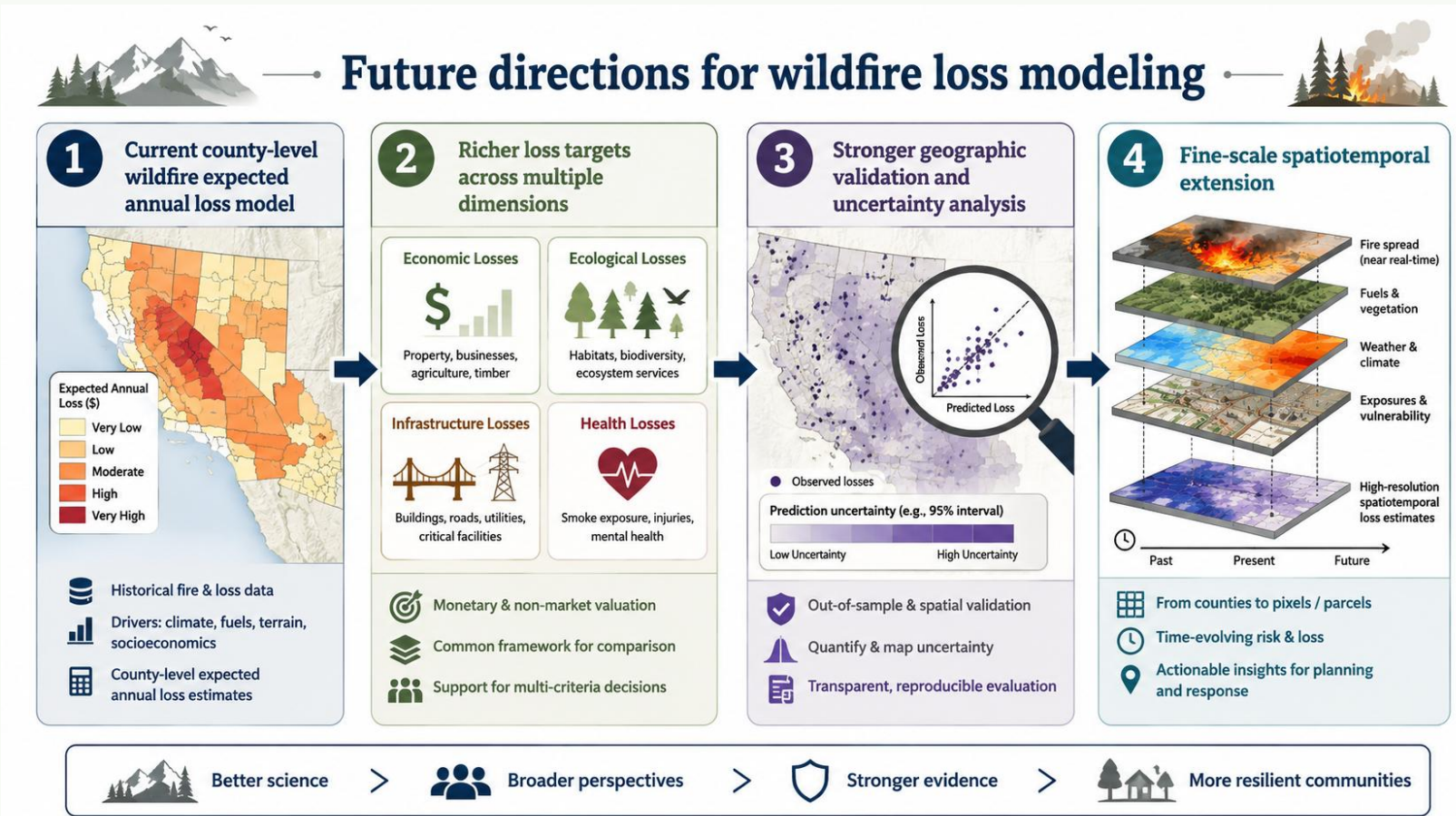
Source: Thesis Chapter 4, Table 4.11; supporting Vietnam conference paper figures.

Limitations and Future Work

The thesis is a defendable baseline, not a final operational loss system.



Limitation	Future extension
County-level EAL is aggregated	Event-level and asset-level losses
CONUS environmental coverage	Broader geographic harmonization
Ecological proxy, not direct loss	Burn severity, ecosystem value, recovery indicators
Grouped validation still imperfect	Ecoregion/climate-zone validation
No uncertainty intervals yet	Calibration and prediction intervals



The main value is a reproducible framework that can be strengthened with richer targets and stricter validation.

Source: Thesis Chapters 5–6, conceptual image illustrates future directions

Conclusion: What the Thesis Demonstrates

Public data can support a meaningful first baseline for consequence-oriented wildfire-loss assessment.

1

Loss-oriented framing

- Moves beyond hazard prediction to monetary wildfire EAL estimation.
- Connects risk prediction to actionable outputs.

2

Environmental signal

- Topography, land cover, and climate substantially improve prediction.
- Grouped validation confirms useful transferability.

3

Broader framework

- Adds ecological screening as a first bridge to ecological consequences.
- Can be adapted to other regions with comparable data.

Research Dissemination and Published Supporting Work

The thesis also documents published supporting work related to wildfire prediction.



Defense relevance

- Supporting An Giang, Vietnam wildfire prediction study was presented at DCHPC 2026.
- The associated conference paper is now published in [IEEE Xplore](#).
- Full title: “**Mapping Forest Fire Risks Using Deep Learning: A Case Study in An Giang Province, Vietnam**”
- Authors: Muhammad Suliman; Ramon Antonio Rodriges Zalipynis
- DOI: 10.1109/DCHPC69296.2026.11517247

Supporting published work aligned with the thesis direction, the FEMA/NRI loss-assessment pipeline remains the core thesis contribution.

Selected References Used in the Slides

References are selected from the thesis literature review and data documentation.

- FEMA. National Risk Index Technical Documentation, 2025.
- Rothermel, R.C. (1972). A mathematical model for predicting fire spread.
- Chuvieco et al. (2010). Development of a framework for fire risk assessment.
- Ager et al. (2010). Wildfire risk and values-at-risk modeling.
- Thompson et al. (2011). Integrated wildfire risk assessment.
- Balch et al. (2017). Human-started wildfires expand the fire niche.

- Abatzoglou (2013). gridMET gridded surface meteorological dataset.
- Dewitz (2021). National Land Cover Database 2019 products.
- Abatzoglou & Williams (2016). Climate change and fuel aridity.
- Williams et al. (2019). Atmospheric aridity and California wildfire activity.
- Roberts et al. (2017). Cross-validation strategies for spatial data.
- Valavi et al. (2019); Meyer & Pebesma (2021). Spatial validation and area of applicability.
- Suliman, M., and R. A. R. Zalipynis. "Mapping Forest Fire Risks Using Deep Learning: A Case Study in An Giang Province, Vietnam." 2026 Fourth International Conference on Distributed Computing and High Performance Computing (DCHPC), IEEE Xplore, 2026.DOI:10.1109/DCHPC69296.2026.11517247.

Thank you for your attention

Questions and Discussion



Project repository and supplementary materials

Muhammad Suliman

Master's Thesis Defense

Economic and Ecological Losses
from Wildfires using Machine
Learning

**Thank you for you attention.
I welcome your questions**

QR code: Project repository and supplementary materials prepared for the thesis defense.

Appendix A: Supporting FEMA/NRI Notebook Outputs

The appendix preserves cleaning, missingness, diagnostic, and component tables without crowding the main story.

What Appendix A documents

- Notebook 1: target construction and missingness.
- Notebook 2: baseline predictors and correlation diagnostics.
- Notebook 4/6: error analysis and grouped-validation residuals.
- Notebook 7: ecological component values behind natural vegetation fraction.

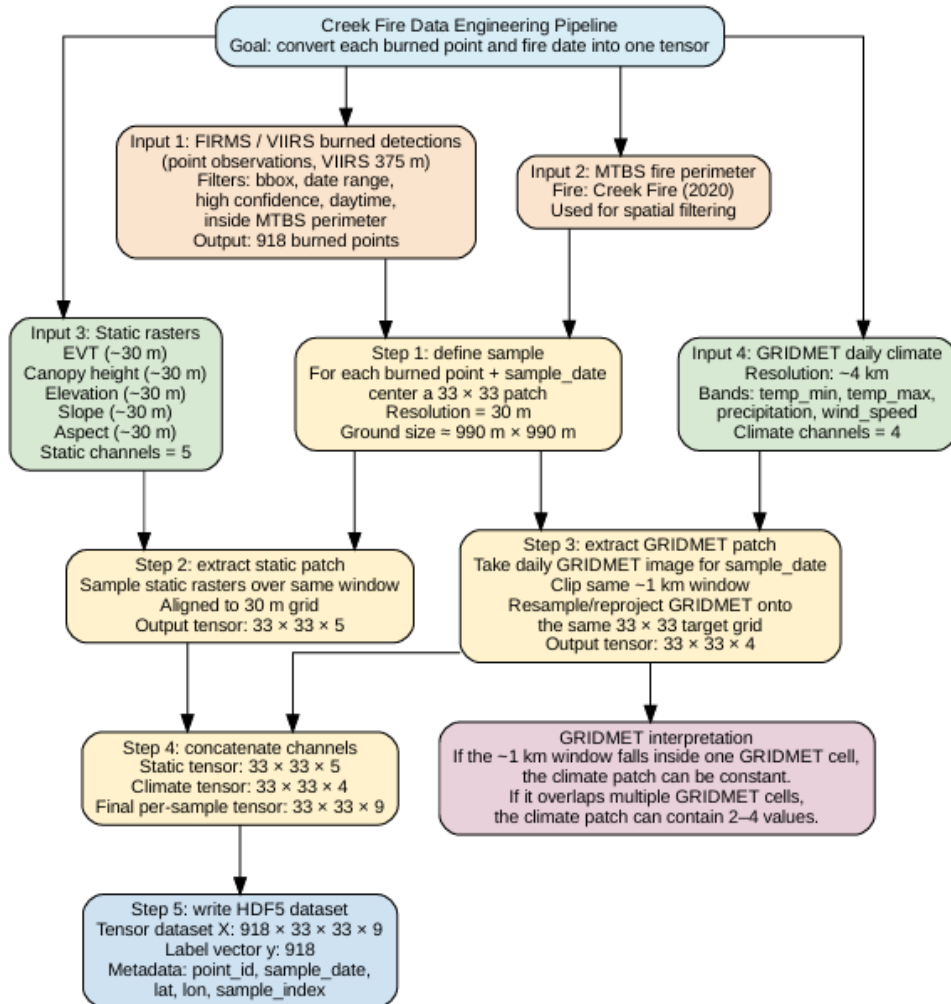
Table A.17: Natural vegetation components for the top ecological-risk counties from Notebook 7.

State	County	FIPS	Forest fraction	Shrub fraction	Grass fraction	Natural vegetation fraction
New Mexico	Lincoln	35027	0.1244	0.5603	0.3057	0.9904
Arizona	Yavapai	04025	0.1579	0.7712	0.0474	0.9765
Nevada	Elko	32007	0.0605	0.6652	0.2458	0.9715
Arizona	Coconino	04005	0.2284	0.6181	0.1226	0.9690
Arizona	Navajo	04017	0.1429	0.7070	0.1263	0.9762
New Mexico	Grant	35017	0.2805	0.4978	0.2037	0.9820

Source: Thesis Appendix A, including Appendix Table A.17.

Appendix B: Creek Fire Fine-Scale Workflow

An earlier spatio-temporal data-engineering workflow supports future fine-scale extensions.



Role in thesis

- Not the main loss-assessment pipeline.
- Demonstrates fire detections, perimeter filtering, static predictors, and climate lag features.
- Useful future bridge from county-level FEMA/NRI modeling to event-level spatial modeling.

Why it is appendix material

- It supports reproducibility and future work, while the main contribution remains county-level monetary wildfire-loss assessment.